

Stock Price Prediction Using Sarima and Prophet Machine Learning Model

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Abstract: *The Prediction of stock market can be defined as to determine the industry stock's future value or other finances which is on the national stock exchange. The advantage of a successful prediction of a stock's future price could results insignificant profit. The efficient-market hypothesis recommends that stock costs mirror all right now accessible data and any value changes that are not founded on recently uncovered data subsequently are an unpredictable. Models like SARIMA and prophet are the machine learning models which are some of best time series machine learning models. Each have its own perspective to choose for a time series machine learning model and use them for the prediction. We will be using both model for prediction of stock price and compute the results and compare them for best accuracy and performance.*

Keywords: Stock price prediction, prophet model, Sarima Model.

I. INTRODUCTION

The time series forecasting has been become a trend in these recent years by the researchers. There are many interesting methods and algorithms, which have been proposed for prediction. Whether one wish to predict some stock prices or some the consumption of electricity, time plays an important factor that is widely considered for the time Series Models. For example, it would be very interesting to know when the price of the stock is going to rise up but also when it is going to rise up.

A time series is simply can be defined as a series of data points which have been ordered in a time orderly manner. In time series, time is usually the independent variable and the goal is usually to make a forecast for the future.

However, there are some aspects which needs to be considered when dealing with time series.[11]

- Is it stationary?
- Is there seasonality?
- Is the target variable autocorrelated?

a) Stationarity

The Stationarity is an important characteristic of the time series. If the statistical properties do not change over the time, the time series can be referred to as stationary. It can be also depicted as that it has constant mean and variance, and the covariance is independent of time. Since we see a growing trend in the stock prices, or the volatility might increase over the time period (meaning the variance is changing).

b) How to test if a process is stationary?

The Dickey-fuller test is used, which is a statistical test that we run to determine if a time series process is stationary or not. The dickey-fuller test is a test which test the null hypothesis that a union root is present.

- If it is, then $p > 0$, and the process is stationary.
- Otherwise, $p = 0$, the null hypothesis is rejected, and the process is considered to be stationary.

c) Seasonality

The periodic fluctuations in the time series is referred as the seasonality. For example, the online sales of a company or website increased during some vacation or festival like Diwali before slowing down again.

The seasonality can also be derived from the autocorrelation plot if it has a sinusoidal shape. The look at the period can give the length of the season.

d) Autocorrelation

The autocorrelation informally can be described as the similarity between observations as a function of the time lag between them. If the autocorrelation plot looks like sinusoidal function, it hints for the seasonality and you can find its value by finding the period in the plot.

II. PREDICTION ALGORITHM

A. Sarima Prediction Model

Seasonal Auto Regressive Integrated Moving Average, SARIMA or Seasonal ARIMA, is an extension of the ARIMA that explicitly supports univariate time series data with a seasonal component.

The SARIMA is formed by including the additional terms in the ARIMA model which was missing in it, which are the seasonal components. The seasonal part of the model consists of the terms that are very similar to the non-seasonal components of the model, but they involve backshifts of the seasonal period.[8]

The AR stands for the autoregressive and refers to be using the lagged values of our target variable to make our prediction. For example, we can use the stock price of today, yesterday's and day before yesterday's prices of stock to forecast the tomorrow's stock price. That would be an AR (3) model as it uses 3 lagged values to make its prediction.

The I stand for integrated. It means that instead we taking the raw values from the target, we are differencing them. For example, our stock price prediction model would try to forecast tomorrow changes in the stock prices. The reason we need this is that many time series exhibit a trend, making the raw values non-stationary and when we take the difference, it makes our y variable more stationary.

The MA stands for the Moving average. A moving average model takes the lagged prediction errors as inputs. This kind of parameter is not directly observable parameter unlike the other parameters and this parameter. At a high level, feeding this model errors back to itself serves to push it somewhat toward the correct values (the actual Y values).

Seasonal Elements

There are four parts of seasonal elements that are not part of ARIMA that must be configured; which are as follows:

- **P:** Seasonal auto regressive order
- **D:** Seasonal Difference order
- **Q:** Seasonal Moving average order
- **m:** No of time steps for a single seasonal period.

A seasonal ARIMA model uses differencing at a lag equal to the quantity of seasons to discard additive/multiplicative seasonality effects. As differencing of lag 1 to remove a trend, the lags differencing defined a moving average term. The seasonal ARIMA model includes autoregressive and moving average terms at lags.[9].

B. Prophet Prediction Model

It is an additive based model, which is a procedure for forecasting time series data in which the non-linear trends are fit with yearly, weekly, and daily seasonality and as well as the holiday effects. The best results of this model are to be seen when the time series have strong seasonal effects and several season of historical data.

The prophet model is vigorous to missing data and the trend shifts and handles the outliers as well. Prophet is an open source software released by Facebook's Core Data Science team. The complex statistical modelling is handled by the stan library and is a prerequisite for prophet.

Prophet follows the sklearn model API. The instance of the Prophet class is been created and then call its fit and predict methods.

The data frame in the prophet model have always two columns: ds and y. The ds (date stamp) column should be a format expected by pandas, it can be of any format like YYYY-MM-DD HH:MM: SS for a timestamp and YYYY-

MM-DD for a date. The y column must be numeric, and it should represent the measurement or attribute which u need to forecast.

How Prophet Works:

As previously said prophet is an additive model which consists of the following components:

$$y(t) = g(t) + s(t) + h(t) + \epsilon$$

- $g(t)$ models' trend, which describes long-term increase or decrease in the data.
- $s(t)$ models' seasonality with Fourier series, which describes how data is affected by seasonal factors such as the time of the year (e.g. more searches for eggnog during the winter holidays)
- $h(t)$ models the impacts of occasions or large events that impact business time series. For example, any new product launch, Diwali, Christmas etc.
- ϵ represents an irreducible error term

Final formula for prophet model

$$\hat{y} = \text{trend} * (1 + \text{multiplicative terms}) + \text{additive terms}$$

Seasonal features
Exogenous regressors
Seasonal features
Exogenous regressors

$$\hat{y}_t = \left[k + \sum_{i=1}^m \delta_i t + m + \sum_{i=1}^m (-\text{change_point}_i \delta_i) \right] * \text{Trend} + \left[1 + \sum_{i=1}^Z \left(\beta_{i,j} \text{ is even} \left(\sin \left(\frac{2\pi t(j+1)}{\text{Period}(i)} \right) \right) + \beta_{i,j} \text{ is odd} \left(\cos \left(\frac{2\pi t(j+1)}{\text{Period}(i)} \right) \right) \right) \right] * \text{Multiplicative terms} + \left[\sum_{i=1}^Z \left(\beta_{i,j} \text{ regressor}(t) - \mu_{\text{train}}(i) \right) \right] + \left[\sum_{i=1}^Z \left(\beta_{i,j} \text{ is even} \left(\sin \left(\frac{2\pi t(j+1)}{\text{Period}(i)} \right) \right) + \beta_{i,j} \text{ is odd} \left(\cos \left(\frac{2\pi t(j+1)}{\text{Period}(i)} \right) \right) \right) \right] + \left[\sum_{i=1}^Z \left(\beta_{i,j} \text{ regressor}(t) - \mu_{\text{train}}(i) \right) \right] * \text{Additive terms}$$

Training Parameters

Prophet model has 5 training parameters:

- k - base trend
- m - offset parameter
- $\delta = \{\delta_i\}$ #change-points=0 - changing of trend in change points
- $\beta = \{\beta_i\}$ $Z_i = 0$ - parameters for Fourier series (seasonal component) and extra regressors (exogenous features), where $Z = 2 * \sum_i$: seasonal-features (Seasonal-order(i)) + # {exogenous regressors}
- σ - level of noise
- mutrain, stdtrain - mean and std value, calculated for each the i -th regressor on train. [10]

III. RESULTS AND DISCUSSION

The experiment was done for 2 machine learning models. The error percentage for both the models (Sarima and Prophet) which are obtained by performing the experiment is given in table 1. From the error percentage its shows that the prophet models perform better than that of another model.

However, as Stock Market is very dynamical system, hence the patterns and dynamics presented by the model may not be same always as stock market data is way more dynamic in real world scenario. This causes to the machine learning model to not able to capture the dynamic change in data points. The Tata Global Beverages stock price dataset has been used for the prediction of the closing price of the stock.

First, the sarima model is been used to train the model and fit the dataset in the sarima function.

	Date	Open	High	Low	Last	Close	Total Trade Quantity	Turnover (Lacs)
0	2018-10-03	208.00	222.25	206.85	216.00	215.15	4642146.0	10062.83
1	2018-10-05	217.00	218.60	205.90	210.25	209.20	3519515.0	7407.06
2	2018-10-04	223.50	227.80	216.15	217.25	218.20	1728786.0	3815.79
3	2018-10-03	230.00	237.50	225.75	226.45	227.60	1708590.0	3960.27
4	2018-10-01	234.55	234.60	221.05	230.30	230.90	1534749.0	3486.05

A sample code of the sarima is shown below:

```
model = auto_arima(training, start_p=1,
start_q=1,max_p=3, max_q=3, m=12,start_P=0,
seasonal=True,d=1, D=1,
trace=True,error_action='ignore',suppress_warning
s=True)

model.fit(training)

forecast = model.predict(n_periods=24)
forecast = pd.DataFrame(forecast,index =
valid.index,columns=['Prediction'])
```

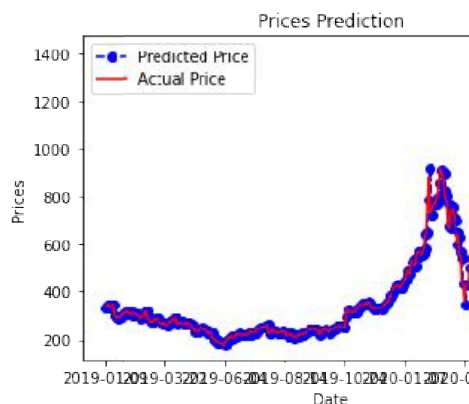
The following pdq values have been used to train the model for the seasonal elements and n_periods dedicate the future values which is going to be predicted by the model which is 24 days.

Results:

The accuracy and result of the model found is depicted as follows using root mean Square error function.

```
rms=np.sqrt(np.mean(np.power((np.array(valid['Close'])-np.array(forecast['Prediction'])),2)))
rms=44.954584993246954
```

Sarima Prediction Chart:



Prophet Model

Sample Code for the Prophet Model is shown below:

```
#fit the model
model = Prophet ()
model.fit(train)
#predictions
close prices =
model.make_future_dataframe(periods=len(valid))
forecast = model. predict(close prices)
```

The model.fit () is used to fit the dataset and train model, and predict the closing price of the stock.

The model. make_future_dataframe() is used for forecasting of the closing price and it takes arguments 'periods' which is no of days which needs to be forecasted.

Another argument which can be given is 'freq' which describes forecasting in days, months or years.

E.g.: freq='D' D stands for days.

freq='M' M stands for Months.

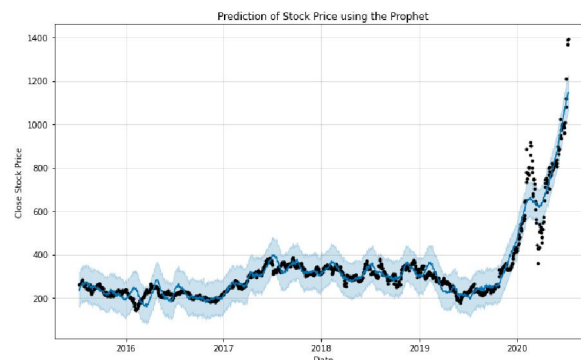
Freq='Y' Y stands for Years.

Results:

The accuracy and result of the model found is depicted as follows using root mean Square error function.

```
#rmse
forecast_valid = forecast['yhat'][987:]
rms=np.sqrt(np.mean(np.power((np.array(valid['y'])-np.array(forecast_valid)),2)))
rms
57.494461930575149
```

Prophet Prediction Chart:



IV. CONCLUSION

As we seen in the results, in time series auto Arima model uses the past data to understand its pattern. These past history values are used by the model to capture the increasing trend in the series. Although the predictions done by this model are far off the real-world stock pricing system, hence these predictions are still not close to the real values. As the plot of the sarima model depicts, the model tends to capture the trend in the time series and forecast the values for future dates.

Like most other time series models, Prophet tends to capture the trend and seasonality from the past data and tries forecasting the future. The Prophet model performs better on various time series datasets, only scenarios like this it gets results off to that of its actual calibre.

Overall, it turns out that the stock prices do not have a seasonality or a particular trend. It mainly depends on the current upcoming in the markets and owing to which the prices of the stock may go high or comes below. Hence, we can depict that techniques like ARIMA, SARIMA and Prophet are not that good of for forecasting of the stock prices and similar kind of problems.

V. FUTURE ENHANCEMENTS

The implementation of this paper can be extended by integrating the technical analysis and fundamental analysis techniques. Through the evaluation of social media analysis particularly on public opinions using fundamental analysis techniques can be incorporated in order to obtain better results. In this way we can provide the improved results for investors in the stock market to choose the better timing for profitable investment decisions.

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